

SAPPORO, Japan

WMO: 474120

Lat: 43.0600N

Lon: 141.3286E

Elev: 86

StdP: 14.65

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.2	16.6	1.4	5.9	18.0	4.5	7.0	19.8	25.9	33.5	21.6	31.1	5.1	170	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.6	85.1	73.0	82.3	71.2	79.8	69.6	74.8	82.5	73.3	80.0	71.8	77.8	8.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	120.2	79.5	71.0	114.4	78.1	69.5	108.8	76.7	38.5	82.8	37.1	80.0	35.7	77.9	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.5	19.4	16.8	10.3	90.4	3.0	3.3	8.1	92.8	6.4	94.7	4.7	96.5	2.5	98.8		
			DB	WB	8.8	76.7	2.9	2.1	6.8	78.2	5.1	79.4	3.5	80.6	1.4	82.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.9	25.7	26.7	33.8	45.4	56.0	63.3	70.9	73.0	65.9	53.9	41.2	30.2	(6)
(7)		DBStd	17.55	4.88	5.49	5.79	6.52	6.02	5.36	4.80	4.65	5.32	5.72	7.18	5.57	(7)
(8)		HDD50	3013	754	651	503	171	16	0	0	0	0	27	276	615	(8)
(9)		HDD65	6425	1219	1071	968	589	286	93	7	2	51	344	715	1080	(9)
(10)		CDD50	2628	0	0	0	31	201	399	647	713	477	149	11	0	(10)
(11)		CDD65	566	0	0	0	0	6	43	189	249	78	1	0	0	(11)
(12)		CDH74	2643	0	0	0	2	56	181	834	1286	283	1	0	0	(12)
(13)		CDH80	526	0	0	0	0	7	23	180	281	35	0	0	0	(13)
(14)	Wind	WSAvg	7.3	6.6	7.2	7.8	8.6	8.5	7.6	7.2	6.9	6.7	7.1	7.0	6.6	(14)
(15)	Precipitation	PrecAvg	45.1	4.3	3.7	3.1	2.4	2.2	2.5	3.2	5.3	5.4	4.5	4.2	4.3	(15)
(16)		PrecMax	65.9	7.2	10.6	6.2	5.2	4.6	6.7	7.6	25.4	12.4	10.1	10.5	8.2	(16)
(17)		PrecMin	28.7	2.0	1.1	1.3	0.2	0.6	0.2	0.6	1.5	1.2	0.7	1.7	2.1	(17)
(18)		PrecStd	6.7	1.3	1.7	1.2	1.3	0.9	1.4	1.7	3.9	2.8	1.9	1.7	1.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.4	44.3	53.9	71.7	80.2	83.1	88.5	90.2	84.0	71.9	62.3	50.4	(19)
(20)		MCWB	36.6	38.5	43.9	53.0	60.4	66.7	75.0	75.8	72.3	60.3	53.1	46.1	(20)	
(21)		2%	DB	37.5	40.4	49.0	64.6	74.2	78.7	84.6	85.9	80.1	68.3	58.7	45.3	(21)
(22)		MCWB	34.0	35.2	40.6	51.0	58.2	64.9	72.8	74.4	68.7	58.7	51.3	40.8	(22)	
(23)		5%	DB	35.2	37.8	45.5	59.5	69.9	75.5	81.6	83.0	77.2	65.6	55.5	41.8	(23)
(24)		MCWB	31.8	33.7	39.0	48.0	55.8	63.7	71.4	72.3	67.5	56.9	49.1	37.5	(24)	
(25)		10%	DB	33.3	35.5	42.7	55.4	65.8	72.3	78.6	80.4	74.6	63.2	52.3	38.6	(25)
(26)		MCWB	30.0	31.7	37.1	46.1	54.0	62.2	69.6	71.0	66.1	55.6	46.8	34.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	39.5	44.9	55.7	62.8	69.3	76.5	77.4	74.2	63.7	55.6	46.5	(27)
(28)		MCDWB	40.1	43.2	51.7	66.7	77.1	79.0	85.3	86.5	81.3	67.7	59.6	49.6	(28)	
(29)		2%	WB	34.6	36.1	42.0	52.1	59.7	66.8	74.3	75.5	72.0	60.8	52.5	41.5	(29)
(30)		MCDWB	36.8	39.2	47.3	62.2	70.5	75.7	82.4	83.4	77.5	66.1	57.4	45.1	(30)	
(31)		5%	WB	32.6	34.1	39.7	49.5	57.6	64.9	72.5	74.1	69.9	58.7	49.7	37.7	(31)
(32)		MCDWB	34.8	37.3	44.6	57.7	66.6	73.0	79.5	80.8	75.0	64.0	54.6	41.0	(32)	
(33)		10%	WB	30.3	32.0	37.5	47.0	55.9	63.3	71.0	72.7	67.4	56.6	47.1	34.8	(33)
(34)		MCDWB	33.0	35.3	42.2	54.2	63.6	70.5	77.0	78.7	72.9	62.1	52.0	38.0	(34)	
(35)	Mean Daily Temperature Range		MDBR	9.5	10.4	11.0	13.5	14.6	13.6	12.2	11.6	13.1	13.6	11.4	9.4	(35)
(36)		5% DB	MCDBR	11.3	13.4	15.5	21.1	22.2	19.0	16.0	14.9	15.2	16.2	16.6	13.4	(36)
(37)		MCWBR	9.9	10.9	10.6	11.5	10.9	9.2	7.6	6.9	8.2	9.8	12.0	11.2	(37)	
(38)		5% WB	MCDBR	11.6	12.9	14.5	18.9	18.5	16.2	14.1	13.0	12.2	14.0	15.6	13.2	(38)
(39)		MCWBR	11.0	11.4	10.9	11.2	10.1	8.5	7.0	6.4	7.8	10.0	12.6	11.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.346	0.397	0.467	0.530	0.547	0.529	0.512	0.477	0.420	0.411	0.393	0.362	(40)	
(41)		taud	1.984	1.840	1.746	1.764	1.823	1.947	2.062	2.177	2.287	2.167	2.113	2.001	(41)	
(42)		Ebn at Noon	233	240	239	236	237	241	244	249	255	237	217	213	(42)	
(43)		Edh at Noon	37	51	64	68	66	58	51	44	37	36	33	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	405	644	1035	1432	1633	1592	1447	1384	1174	832	469	331	(44)	
(45)		RadStd	29	55	86	149	143	147	161	106	108	49	48	27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page